

# Entitlement Lifecycle, Value, Consumption & Intelligence

Portfolio, Ownership & Entitlement Management —  
Functional Specification

# Table of Contents

|                                                                         |    |
|-------------------------------------------------------------------------|----|
| Board 1 — Portfolio, Ownership & Entitlement Management.....            | 3  |
| Screen 1 — Portfolio Command Center.....                                | 3  |
| Screen 2 — Portfolio Profile & Structure.....                           | 4  |
| Screen 3 — Family & Dependent Hierarchy.....                            | 4  |
| Screen 4 — Corporate, School & Group Portfolio Manager.....             | 5  |
| Screen 5 — Unified Entitlement Inventory.....                           | 6  |
| Screen 6 — Entitlement Assignment Manager.....                          | 7  |
| Screen 7 — Ownership, Transfer & Sharing Center.....                    | 8  |
| Screen 8 — Portfolio Permissions & Spending Controls.....               | 9  |
| Screen 9 — Credential & Media Association.....                          | 10 |
| Screen 10 — Portfolio 360 & Relationship Map.....                       | 10 |
| Board 1 — Integration Architecture.....                                 | 12 |
| Board 2 — Entitlement Lifecycle, Value, Consumption & Intelligence..... | 13 |
| Screen 1 — Entitlement Lifecycle Command Center.....                    | 14 |
| Screen 2 — Entitlement Lifecycle & Status Manager.....                  | 15 |
| Screen 3 — Reservation, Hold & Release Manager.....                     | 16 |
| Screen 4 — Real-Time Consumption Monitor.....                           | 17 |
| Screen 5 — Usage Rules & Consumption Controls.....                      | 18 |
| Screen 6 — Upgrade, Exchange & Conversion Center.....                   | 19 |
| Screen 7 — Refund, Credit & Value Recovery.....                         | 20 |
| Screen 8 — Expiry, Renewal & Notification Center.....                   | 21 |
| Screen 9 — Entitlement Intelligence & Optimization.....                 | 22 |
| Screen 10 — Entitlement Audit, API & Integration Center.....            | 23 |

# Board 1 — Portfolio, Ownership & Entitlement Management

**Objective:** Provide the central backend workspace for creating and managing guest portfolios, family/dependent structures, organizational portfolios, ownership, entitlement assignment, transfers, sharing, and associated ticket/credential media.

This board primarily addresses **5.5.1–5.5.19 and related 5.5 requirements**, without rebuilding the underlying Ticketing, Wallet, Loyalty, Membership, CRM, Payment, or Access Control engines.

## Screen 1 — Portfolio Command Center

Central operational dashboard for all portfolios.

### Functions:

- Search by guest name, email, mobile, account ID, portfolio ID, membership, ticket, QR, RFID, wallet or loyalty ID.

- View Individual, Family, Corporate, School, Tour Group and Organization portfolios.
- Display portfolio owner and number of linked members.
- Show active tickets, memberships, passes and entitlements.
- Show wallet/credit and loyalty summary references.
- Flag expiring, suspended, transferred or problematic entitlements.
- Filters by venue, portfolio type, status, organization and date.
- Quick actions: View, Assign, Transfer, Share, Merge and Suspend.
- AI highlights unusual activity and portfolios requiring attention.

**Key KPI cards:** Total Portfolios | Active Entitlements | Expiring Soon | Pending Transfers | Exceptions.

---

## Screen 2 — Portfolio Profile & Structure

Detailed configuration of an individual portfolio.

### Functions:

- Portfolio ID and name.
- Portfolio type.
- Primary account holder.
- Linked guest/customer profile.
- Preferred venue/brand.
- Status and effective dates.
- Default language and communication preferences.
- Portfolio-level restrictions.
- Tags and internal notes.
- View consolidated assets owned by the portfolio.

The screen should clearly visualize:

## Screen 3 — Family & Dependent Hierarchy

Manage family structures and dependent portfolios.

### **Functions:**

- Primary account holder.
- Spouse/guardian relationships.
- Children/dependents.
- Relationship type.
- Age/date-of-birth restrictions.
- Guardian permissions.
- Shared versus individual assets.
- Allow primary holder to manage dependent tickets and memberships.
- Shared family wallet reference.
- Shared/personal loyalty configuration.
- Move entitlement between eligible family members.
- Parental/guardian approval requirements.
- Family hierarchy visualization.

Example:

### **Qossai — Primary Account**

↳ Omar — Child Portfolio

↳ Rima — Child Portfolio

↳ Member 3 — Dependent Portfolio

Each remains individually traceable even when assets or balances are shared.

---

## Screen 4 — Corporate, School & Group Portfolio Manager

Manage non-family portfolio structures.

Supports:

- Companies
- Schools
- Tour operators
- Travel agencies
- Event organizers
- Corporate groups
- Employee programs
- Contractors/visitors

**Functions:**

- Organization master account.
- Group administrators.
- Participants/members.
- Bulk member import.
- Allocate tickets/passes/entitlements.
- Employee/student identifiers.
- Group-level versus individual benefits.
- Spending limits.
- Transfer permissions.
- Validity periods.
- Bulk assignment/revocation.
- Group utilization.

Example:

**Greenfield School — 120 Guests**

- 110 Student Portfolios
  - 10 Staff Portfolios
  - 120 Admission Entitlements
  - 120 Meal Entitlements
  - 3 Group-Leader Credentials
- 

## Screen 5 — Unified Entitlement Inventory

This is one of the most important screens.

It provides a consolidated view of everything a portfolio/member is entitled to use.

Supported entitlement categories include:

**Tickets | Memberships | Annual Passes | Attractions | Experiences | Events | F&B | Retail Benefits | Parking | Lockers | Meal Plans | Vouchers | Packages | Fast Pass | Other Benefits**

For every entitlement display:

- Entitlement ID.
- Product/source.
- Owner.
- Assigned user.
- Quantity/remaining uses.
- Validity.
- Reservation requirement.
- Consumption status.
- Credential/media.
- Transferability.
- Upgrade eligibility.
- Current status.

Statuses include:

**Active | Reserved | Partially Consumed | Consumed | Expired |  
Suspended | Refunded | Transferred | Cancelled**

---

## Screen 6 — Entitlement Assignment Manager

Controls who receives and can consume an entitlement.

### **Functions:**

- Assign purchased entitlement to portfolio member.
- Bulk assignment.
- Automatic assignment rules.
- Manual reassignment.
- Assign unallocated tickets.
- Assign package components independently where permitted.
- Validate eligibility before assignment.
- Age restrictions.
- Membership requirements.
- Guest-type requirements.
- Waiver requirements.
- Assignment effective dates.
- Approval workflow for controlled assets.
- Assignment history.

AI can recommend the most appropriate portfolio member when assignment is unclear.

---

## Screen 7 — Ownership, Transfer & Sharing Center

Manage movement of assets between eligible portfolios.

Supports transfer of:

- Tickets
- Memberships where allowed
- Experiences
- Packages
- Vouchers
- Promotional credits
- Stored value
- Benefits
- Other transferable entitlements

**Transfer configuration:**

- Transfer allowed/not allowed.
- Same-family only.
- Linked accounts only.
- Named guest restrictions.
- Number of transfers allowed.
- Transfer deadline.
- Transfer fee.
- Approval requirement.
- Recipient eligibility.
- Temporary versus permanent transfer.

Every transfer must retain the complete chain:

**Original Owner → Transfer → Recipient → Consumption**

No transaction history should be lost.

---

## Screen 8 — Portfolio Permissions & Spending Controls

Configure who is allowed to manage or consume assets.

Examples:

A parent may allow a child to:

- Use admission ticket: **Yes**
- Spend wallet balance: **Up to AED 100/day**
- Redeem loyalty points: **No**
- Transfer ticket: **No**
- Purchase F&B: **Yes**
- Purchase Retail: **AED 200/day**
- Upgrade ticket: **Requires approval**

Corporate accounts can similarly establish employee/group limits.

Controls should support:

- Asset management permission.
  - Spending limit.
  - Transfer permission.
  - Assignment permission.
  - Purchase category restrictions.
  - Age restrictions.
  - Venue restrictions.
  - Time restrictions.
  - Approval thresholds.
- 

## Screen 9 — Credential & Media Association

Connect portfolio entitlements to the credential used by the guest.

Supported media:

**Dynamic QR | Static QR where permitted | RFID Wristband | RFID Card |  
NFC | Mobile Credential | Digital Ticket | Printed Ticket**

Functions:

- View active credential.
- Link credential to portfolio/member.
- View entitlements accessible through credential.
- Replace lost media.
- Reissue credential.
- Suspend compromised media.
- Credential history.
- Multiple-media policy.
- Device association where applicable.
- Integration with Access Control.

Importantly, **this screen does not generate Access Control rules**. Those remain inside the Access Control module we already designed.

It only manages the **portfolio ↔ credential ↔ entitlement relationship**.

---

## Screen 10 — Portfolio 360 & Relationship Map

The final screen gives management a graphical consolidated view.

Example:

### **Guest Account**

- Family Portfolio
- 4 Members

### **Member A**

- Annual Pass
- Wallet
- Loyalty
- Parking

### **Member B**

- Admission Ticket

- Fast Pass
- Meal Entitlement
- RFID Wristband

### **Member C**

- Membership
- Experience Reservation
- QR Credential

Users can click any relationship to open its authoritative module.

The screen should also display:

- Ownership relationships.
- Shared assets.
- Current assignments.
- Pending transfers.
- Expiring entitlements.
- Restrictions.
- Consumption summary.
- Portfolio health.
- AI recommendations.

### **AI examples:**

3 unused entitlements expire within 14 days.

Family portfolio has AED 320 promotional credit expiring this month.

Two child portfolios have overlapping attraction entitlements.

Recommend transferring unused entitlement to eligible family member.

Annual Pass upgrade may provide better value based on current family utilization.

---

# Board 1 — Integration Architecture

This board should orchestrate existing TICVAI engines rather than duplicate them:

## Portfolio & Entitlement

- ↔ Customer 360 / CRM
- ↔ Ticketing Catalogue
- ↔ Membership
- ↔ Loyalty
- ↔ Wallet
- ↔ Payment
- ↔ Access Control
- ↔ Reservations
- ↔ F&B
- ↔ Retail
- ↔ Notifications
- ↔ Reporting & AI
- ↔ Audit & Approval

Every assignment, transfer, ownership change, credential association and administrative override must be **fully auditable**.

## Board 1 acceptance outcome

After implementation, an authorized operator should be able to answer one question from a single workspace:

**“Who owns what, who is allowed to use it, how can it be transferred/shared, and what credential is associated with it?”**

That is the purpose of Board 1.

# Board 2 — Entitlement Lifecycle, Value, Consumption & Intelligence

**Module:** Portfolio & Entitlement Management

**Board:** 2 of 2

**Screens:** 10

**Primary Matrix Coverage:** 5.5.9–5.5.30, complementing Board 1's ownership, hierarchy, assignment and credential capabilities.



**Board Objective**

Board 1 was answered:

**Who owns the entitlement, who can use it, and how is it assigned or transferred?**

Board 2 answers:

**What happens to the entitlement from issuance until consumption, expiry, refund or cancellation—and how do we control, monitor and optimize it?**

This board must act as an **orchestration layer**. Ticketing, Wallet, Loyalty, Membership, Payments and Access Control remain their respective authoritative engines.

---

## Screen 1 — Entitlement Lifecycle Command Center

Central dashboard for monitoring the complete entitlement lifecycle across TICVAI.

### Core functions

Display entitlements by lifecycle status:

- Active
- Reserved
- Partially Consumed
- Consumed
- Expiring Soon
- Expired
- Suspended
- Refunded
- Transferred
- Cancelled

Provide KPI cards:

**Total Active | Reserved | Expiring Soon | Suspended | Refund Pending | Exceptions**

Filters:

- Venue
- Product
- Entitlement type
- Portfolio

- Guest
- Status
- Validity period
- Channel
- Membership
- Source transaction

Allow authorized operators to drill from portfolio → entitlement → transaction  
→ consumption history.

### **AI capability**

AI identifies abnormal lifecycle conditions such as:

437 entitlements expire within 7 days with no reservation.

Unusually high refund activity detected for Product X.

128 annual-pass benefits remain unused despite high visit frequency.

---

## **Screen 2 — Entitlement Lifecycle & Status Manager**

Configure and manage the state machine controlling each entitlement.

### **Lifecycle model**

**Issued → Active → Reserved → Consumed**

with controlled alternative paths:

**Active → Suspended**

**Active → Transferred**

**Active → Refunded**

**Active → Expired**

**Reserved → Released**

**Reserved → No-show**

**Any eligible state → Cancelled**

## Configuration

Administrators configure:

- Allowed status transitions.
- Automatic transitions.
- Manual transitions.
- Effective date/time.
- Suspension reasons.
- Cancellation reasons.
- Refund triggers.
- Expiry behavior.
- Reinstatement rules.
- Required approvals.
- Notification triggers.

Every manual lifecycle override requires:

**User + Timestamp + Reason + Previous State + New State**

and is written to the immutable audit history.

---

## Screen 3 — Reservation, Hold & Release Manager

Manage entitlement reservations before actual consumption.

This directly supports the matrix requirement to **reserve entitlements before use and automatically release them when unused.**

### Functions

- Reserve an entitlement.
- Select attraction/experience/service.
- Select date/session/timeslot.
- Configure reservation validity.

- Set hold duration.
- Set release deadline.
- Configure grace period.
- Automatic release.
- Manual release.
- Extend reservation.
- Change reservation.
- Waitlist integration.
- Capacity synchronization.

Example:

Fast Pass

Attraction: Formula Ride

Reservation: 14:30

Grace Period: 15 minutes

Auto Release: 14:45

Released capacity becomes immediately available to the appropriate inventory/capacity engine.

---

## Screen 4 — Real-Time Consumption Monitor

One of the most important operational screens.

Shows **where, when and how an entitlement is actually being consumed.**

### Sources

Consumption may originate from:

- Attraction
- Access gate
- Event
- F&B POS

- Retail POS
- Parking
- Locker
- Experience
- Mobile application
- Kiosk
- External integrated system

### **Information**

For each consumption event:

- Guest
- Portfolio
- Entitlement
- Location
- Device
- Credential
- Date/time
- Quantity
- Previous balance
- Consumed amount
- Remaining balance/uses
- Validation result

Example:

### **Meal Plan — 5 Meals**

5 → 4 → 3 → 2 → 1 → Consumed

Updates should occur in **real time across all connected channels**.

## Screen 5 — Usage Rules & Consumption Controls

Controls **how an entitlement may be consumed**.

### Rule configuration

Support:

- Single use.
- Multiple use.
- Unlimited use.
- Quantity-based consumption.
- Value-based consumption.
- Daily limits.
- Weekly limits.
- Per-attraction limits.
- Per-session limits.
- Time restrictions.
- Venue restrictions.
- Location restrictions.
- Guest restrictions.
- Age restrictions.
- Membership restrictions.
- Simultaneous-use prevention.

Example:

### Gold Fast Pass

Unlimited eligible attractions per day

BUT

Maximum **1 use per ride every 60 minutes**

The configuration should call the appropriate validation/access engines rather than duplicate Access Control logic.

---

## Screen 6 — Upgrade, Exchange & Conversion Center

Central workspace for changing an existing entitlement.

Supports matrix requirements for upgrades and exchanges.

### **Upgrade examples**

**General Admission → Premium Admission**

**Day Ticket → Annual Pass**

**Standard Membership → Gold Membership**

**Standard Experience → VIP Experience**

### **Exchange examples**

**10:00 Session → 14:00 Session**

**Experience A → Experience B**

### **Functions**

- Search original entitlement.
- Determine eligibility.
- Calculate price difference.
- Calculate applicable fees.
- Preserve consumed value.
- Preserve original transaction reference.
- Issue replacement entitlement.
- Invalidate old entitlement where required.
- Update credential.
- Update portfolio.
- Update capacity.
- Update financial records.

Full before/after history must remain available.

---

## Screen 7 — Refund, Credit & Value Recovery

Manage financial consequences when an entitlement is partially or fully returned.

### Supported actions

- Full refund.
- Partial refund.
- Wallet credit.
- Account credit.
- Promotional credit.
- Voucher issuance.
- Original payment-method refund.
- Mixed refund where supported.

### Controls

System evaluates:

- Amount originally paid.
- Already-consumed value.
- Non-refundable components.
- Cancellation fee.
- Refund fee.
- Promotional contribution.
- Wallet contribution.
- Loyalty redemption.
- Taxes/fees.
- Remaining refundable amount.

Example:

Package Price: AED 500  
Consumed Value: AED 150  
Cancellation Fee: AED 25  
**Refundable: AED 325**

Once approved, TICVAI synchronizes the outcome with **Payment, Finance, Wallet, Portfolio, Capacity and entitlement status** as applicable.

---

## Screen 8 — Expiry, Renewal & Notification Center

Central configuration for time-sensitive portfolio assets.

Monitor:

- Tickets
- Memberships
- Annual passes
- Wallet credit
- Promotional credit
- Gift cards
- Vouchers
- Loyalty points
- Experiences
- Benefits

### Configuration

Administrators define:

- Valid from.
- Valid until.
- Rolling validity.
- Activation-based validity.
- Grace period.

- Renewal eligibility.
- Auto-renewal where applicable.
- Expiry warning schedule.
- Expiry behavior.
- Extension permissions.

## **Communication**

Notifications through configured:

**Email | SMS | WhatsApp | Push | In-App**

Example:

Your AED 120 promotional credit expires in 7 days.

AI can determine the best engagement action rather than simply issuing a generic expiry warning.

---

## **Screen 9 — Entitlement Intelligence & Optimization**

This is the main **AI screen for Board 2**.

AI analyzes portfolio behavior across:

**Ticketing + Membership + Loyalty + Wallet + F&B + Retail + Reservations + Visits + Entitlement Consumption**

**AI insights**

**Expiry risk**

14,820 entitlements expire within 30 days.

Predicted unused: 3,420.

**Upgrade opportunity**

Guest visited 4 times using day tickets.

Annual Pass would have reduced expected annual cost by AED 280.

**Underutilization**

Premium Membership benefit utilization: 31%.

## **Cross-sell**

Families purchasing Attraction Package A show high affinity for Meal Package B.

## **Portfolio optimization**

Three family members maintain separate promotional balances. Consolidation is permitted and may reduce expiration risk.

## **Abuse/fraud signals**

Unusual entitlement transfers detected between unrelated portfolios.

AI recommendations remain advisory unless the client explicitly configures an approved automation policy.

---

# **Screen 10 — Entitlement Audit, API & Integration Center**

This screen provides the governance and technical control layer.

## **Audit timeline**

Record:

- Creation
- Assignment
- Reassignment
- Reservation
- Release
- Transfer
- Sharing
- Consumption
- Upgrade
- Exchange
- Refund
- Suspension

- Expiration
- Cancellation
- Credential changes
- Administrative overrides

Each record includes:

**Who | What | When | Where | Previous Value | New Value | Reason | Source System**

### **Integration monitoring**

Show synchronization status with:

- Ticketing
- CRM
- Membership
- Loyalty
- Wallet
- Payment
- Access Control
- F&B
- Retail
- Reservation
- Mobile App
- External APIs

### **APIs**

Secure APIs should support authorized:

**Create | Retrieve | Update | Assign | Reserve | Release | Transfer | Consume | Suspend | Cancel**

with authentication, authorization, idempotency, rate limits and audit logging.

## **Cross-Board Relationship**

The two Portfolio & Entitlement boards now have a very clean separation:

### **Board 1**

#### **Portfolio, Ownership & Entitlement Management**

##### **WHO owns WHAT?**

Account → Portfolio → Member → Entitlement → Credential

### **Board 2**

#### **Entitlement Lifecycle, Value, Consumption & Intelligence**

##### **WHAT happens to it?**

Issue → Reserve → Consume → Upgrade/Exchange → Refund → Expire/Cancel

This avoids duplicating functionality while providing with the full operational picture.

---

### **Board 2 Acceptance Criteria**

Board 2 will be accepted when TICVAI can:

- Track an entitlement throughout its complete lifecycle.
- Reserve and automatically release unused entitlements.
- Track real-time consumption across connected business areas.
- Enforce configurable consumption restrictions.
- Handle upgrades and exchanges while preserving history.
- Process partial/full refunds with financial reconciliation.
- Track expiry and trigger communications.
- Provide AI-driven utilization, expiry, upgrade and cross-sell recommendations.
- Maintain immutable audit history.
- Synchronize entitlement state across all relevant TICVAI engines.
- Expose secure APIs for external and internal integrations.

Together, **Boards 1 + 2 complete the Portfolio & Entitlement Management area (5.5)** without rebuilding Ticketing, Wallet, Membership, Loyalty, CRM, Payment or Access Control.